Experiment 13

Case Study on Limitations of Low-Code Platforms for a Complex Enterprise Use Case

Name: Midhun J| Register Number: 241401055

Aim

To studythe limitations of low-code platforms when they are applied to complex enterprise-level applications and to understand when pro-code extensions or hybrid approaches may become necessary.

Introduction

Low-codeplatforms allow applications to be developed mainly through visual development tools instead of writing large amounts of traditional program code. Applications are assembled using predefined components such as forms, reports and dashboards, and behaviour is defined through workflows, rules and configuration. A small amount of scripting is usually available for specific needs. This approach reduces development time and effort and makes it possible for a wider range of people to take part in building business applications. Low-code platforms are well suited to standard business applications. As the requirements of an enterprise grow in terms of users, data volume, process complexity, integrations, security and customisation, the built-in capabilities of a platform may no longer be sufficient on their own. This case study examines such limitations through an enterprise job application and recruitment scenario, which extends the Job Application Form application built in Zoho Creator.

241401055

Case Study: Enterprise Job Application Management System

Anenterpriseorganisationrequires a systemtomanageitsrecruitmentoperations.Thesystem may need to serve the following users and handle the following functions:

Category	***Items***
Users	Applicants, recruiters / HR executives, hiring managers, interviewers and administrators
Data and operations	Applicant records, job applications, application screening, interview scheduling and application history
Supporting functions	Notifications, approvals and resume / document storage
Enterprise integration and insight	CRM integration, HRMS integration, analytics and reporting

Figure 1 shows the architecture of this system on a low-code platform. The architecture is arranged as six groups that are connected from left to right:

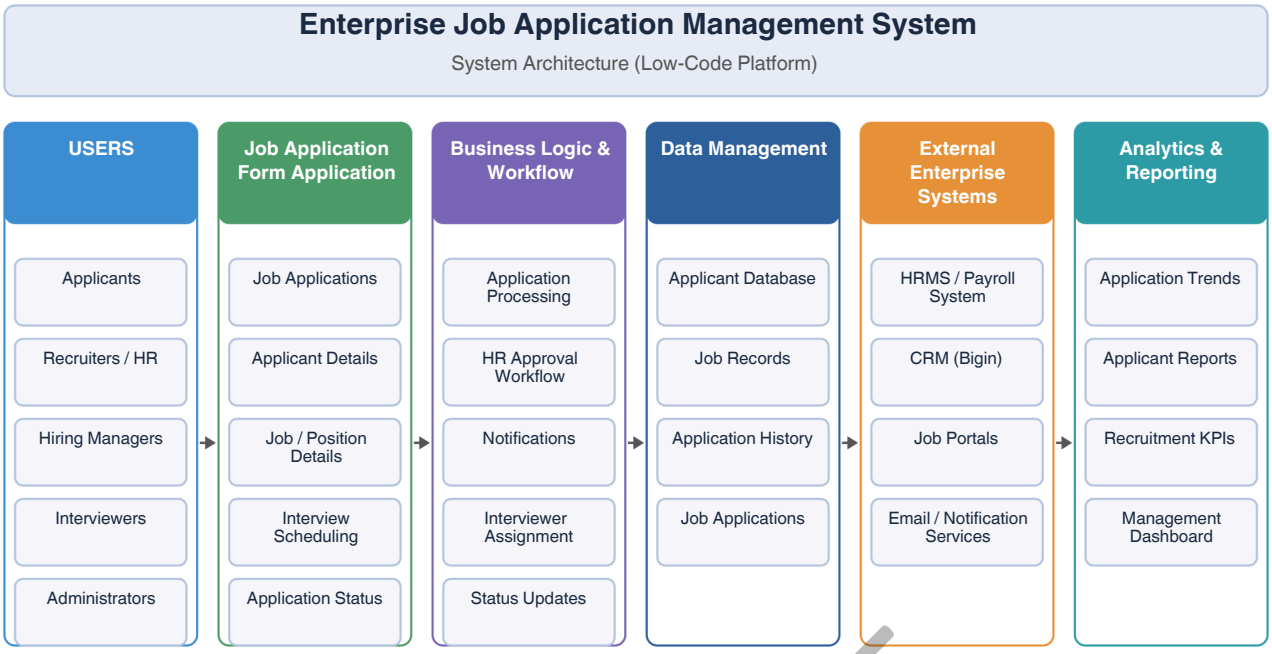


Figure 1: Enterprise Job Application Management System Architecture

- ***Users:*** applicants,recruiters/HRexecutives,hiringmanagers,interviewersandadministrators, who interactwiththe system.
- ***Job Application Form Application:*** theuser-facingpart,coveringjobapplications,applicant details, job/positiondetails,interviewschedulingandapplicationstatus.
- ***Business Logic & Workflow:*** applicationprocessing,HRapprovalworkflow,notifications, interviewer assignment andstatusupdates,which moveanapplicationthroughtheprocess.
- ***Data Management:*** the applicant database,jobrecords,applicationhistoryandjobapplications, in whichthe informationis stored.
- ***External Enterprise Systems:*** the HRMS/payrollsystem,the CRM(Bigin),jobportals and email / notification serviceswith whichthe applicationmustexchangeinformation.
- ***Analytics & Reporting:*** application trends,applicant reports, recruitmentKPIsanda management dashboard, which present thestored and processed data todecision makers.

The groups interact in sequence. Users enter and view information through the application; the business logic and workflow layer processes this information; the data layer stores and supplies it; external systems provide or receive related information such as customer relationships, payroll and job postings; and the analytics layer reports on the results.

24140155

How a Low-Code Platform Can Support the System

A large part of the system consists of standard application functionality that low-code platforms are designed to handle:

Capability	*Use in the Job Application System*
Forms	Capturing job applications, applicant details and job details.
Reports	Viewing job applications, application history and applicant reports.
Workflows	Moving an application through processing, HR approval, assignment and status updates.
Business process automation	Applying rules and automatic actions, such as updating a status when a step is completed.

Capability	*Use in the Job Application System*
Data management	Storing applicant, job and application records and performing standard create, read, update and delete operations.
User interfaces	Providing web and mobile screens for applicants, recruiters, managers and administrators.
Notifications	Informing applicants about application status through email or similar messages.
Basic integrations	Connecting to common services such as a CRM through built-in connectors or standard interfaces.

Limitations at Enterprise Scale

As the system grows into a complex enterprise application, the following limitations may become significant. They are discussed individually below.

Performance and Scalability

When the number of job applications, applicant records and users becomes very large, performance may be affected. Complex queries and reports over large amounts of data can respond slowly, and the platform may impose limits on how far an application can be scaled. Because the underlying infrastructure is managed by the platform, the options for tuning performance are limited.

Complex Business Logic

Highly specialised business rules, such as intricate multi-level approval chains, candidate scoring and shortlisting across many positions, and advanced processing, may be difficult to express using visual workflows and standard scripting. Such requirements may need custom development.

System Integration

Connecting the application to HRMS and ERP systems, CRM systems, job portals, legacy systems, external APIs and other enterprise services can be challenging. Built-in connectors may not cover every system or data format, and complex or real-time integrations may need custom interfaces.

Customization

Standard low-code components provide common screens and layouts. Highly specialised interfaces and user experiences, or functionality that is not offered by the platform, may exceed what these standard components can provide.

Security and Governance

Enterprises require detailed access control, governance, compliance and strong security practices. Role- and permission-based controls offered by a platform may need to be supplemented to meet fine-grained, regulatory or organisation-specific requirements, such as the protection of applicant personal data, and governance of many applications must be managed carefully.

AI and Advanced Technology

Specialised artificial intelligence and machine-learning models, such as screening resumes or ranking candidates, and other advanced processing may require external services or custom code, since a low-code platform may not provide such capabilities directly.

Vendor / Platform Dependency

The application depends on the capabilities, restrictions, pricing and supported integrations of the selected platform. Features may be limited by the plan or edition, and moving an application to another platform later can be difficult.

24140105

Data Management

Large datasets, complex relationships between records and advanced data processing, such as heavy batch processing or complex data transformations, may be difficult to handle within the data model and tools provided by the platform.

Figure 2 summarises how the limitations arise. A basic low-code application, characterised by quick development, minimal configuration and suitability for small and medium needs, faces growing enterprise requirements: more users and data, complex business processes, integration with multiple systems and higher security and compliance needs. These lead to enterprise-scale challenges such as increased complexity, higher performance demands, greater integration needs, stronger security and governance, and more customisation and flexibility. The diagram then identifies eight key limitations, which are the ones discussed above.

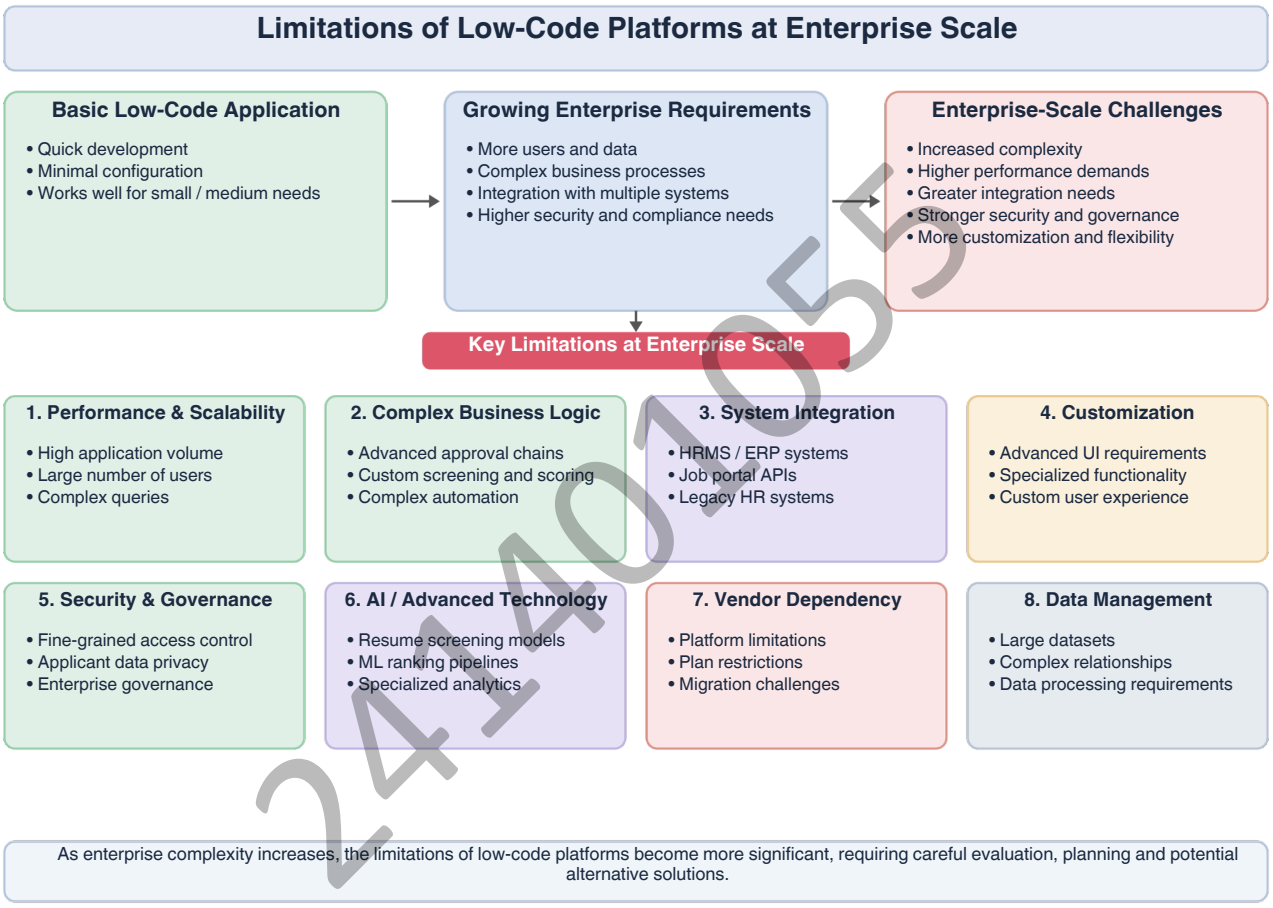


Figure 2: Limitations of Low-Code Platforms at Enterprise Scale

The relevance of each limitation to the Enterprise Job Application Management System is given below.

Limitation	***Relevance to the Case Study***
Performance & Scalability	Growing numbers of applicants, job applications and users, and reports over application history.
Complex Business Logic	Approval workflows, interviewer assignment and screening rules that become more complex over time.
System Integration	CRM (Bigin), HRMS / payroll system, job portals and email / notification services.
Customization	Specialised interfaces for recruiters, interviewers and managers, and a tailored applicant experience.

24140155

Limitation	*Relevance to the Case Study*
Security & Governance	Different access needs for applicants, recruiters, managers and administrators; protection of applicant personal data.
AI / Advanced Technology	Possible intelligent features, such as resume screening, built on application history and recruitment data.
Vendor Dependency	Reliance on the platform for all application, workflow and reporting features, and on its plan limits.
Data Management	Large volumes of applicant records and application history with relationships to jobs and approvals.

Low-Code vs Pro-Code Requirements

The suitability of low-code or pro-code development depends on the requirements of the application. Neither approach is universally better; the comparison below indicates the typical role of each.

Area	*Low-Code Approach*	*When Pro-Code May Be Required*
User interface	Standard screens, forms and layouts built from predefined components.	Highly specialised interfaces or custom user experiences beyond standard components.
Business logic	Visual workflows, rules and simple scripting for common business processes.	Complex rules and advanced processing that cannot be expressed in visual tools.
Data management	Built-in data model with forms, relationships and standard create, read, update and delete operations.	Very large datasets, complex relationships and advanced data processing.
Integrations	Built-in connectors and standard interfaces for common services.	Legacy systems, specialised or real-time integrations and systems without connectors.
Automation	Rules, scheduled actions and workflow-based automation.	Complex or high-volume automation and specialised processing.
AI/ML	Basic features offered by the platform, where available.	Custom AI models, machine-learning pipelines and specialised analytics.
Security	Role- and permission-based access control provided by the platform.	Fine-grained access control, specific compliance requirements and enterprise security needs beyond platform features.
Scalability	Suitable where the platform handles the expected users and data volumes.	High data volumes, large user numbers and demanding performance requirements.
Custom functionality	Functionality available as platform features.	Functionality that the platform does not provide.

24140105

Hybrid Approach

Organisationscancombine low-code development with pro-code extensions. Low-code is used to develop standard application functionality quickly, while custom code, APIs, integrations or specialised processing are used for requirements that exceed the built-in capabilities of the platform. The structure consists of three layers:

Low-Code Application → Pro-Code Extensions → Enterprise Systems

Figure 3 shows this architecture. Users such as applicants, recruiters, hiring managers, interviewers and administrators access the system at the top, and the three layers are described below.

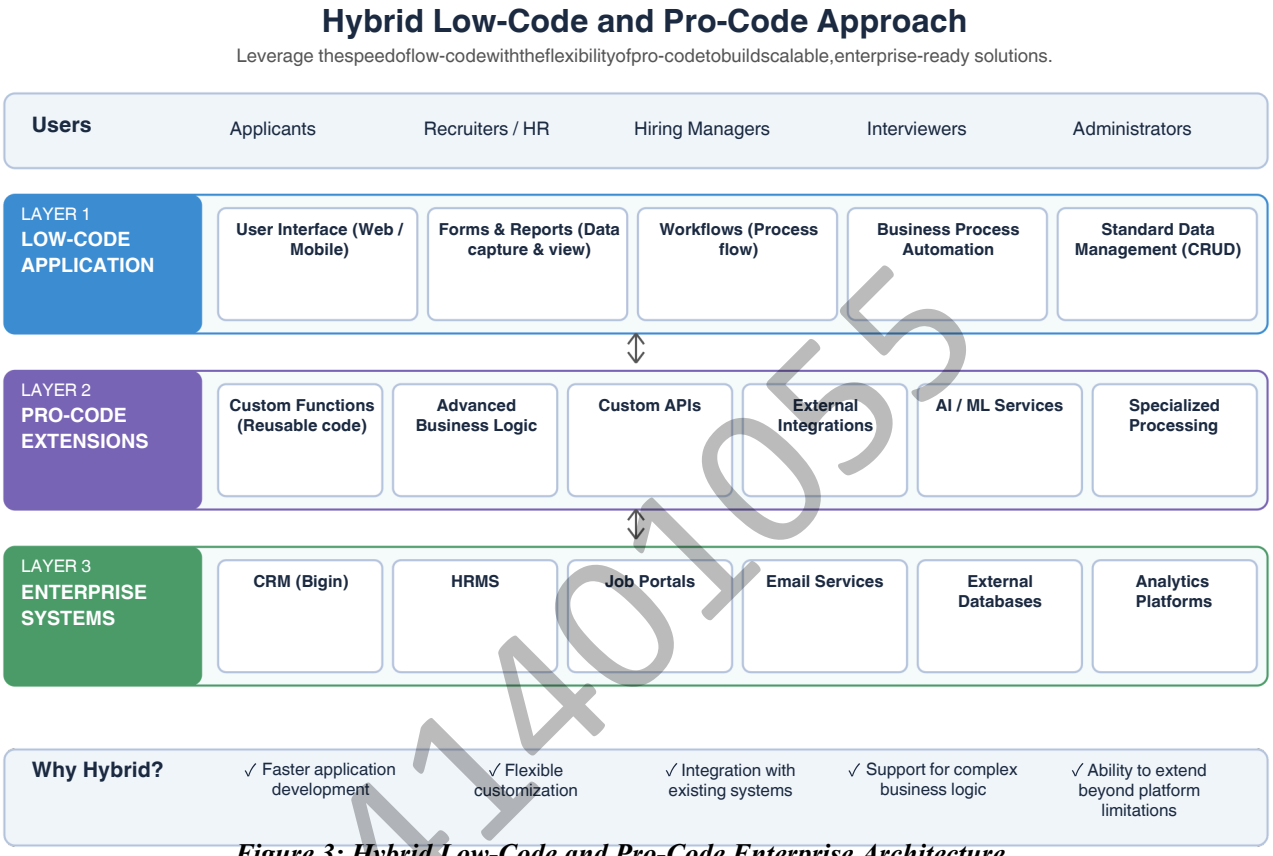


Figure 3: Hybrid Low-Code and Pro-Code Enterprise Architecture

Layer	*Role*
Layer 1 – Low-Code Application	Provides rapid development for standard application needs: user interface (web / mobile), forms and reports, workflows, business process automation and standard data management.
Layer 2 – Pro-Code Extensions	Adds custom logic and specialised capabilities: custom functions, advanced business logic, custom APIs, external integrations, AI / machine-learning services and specialised processing.
Layer 3 – Enterprise Systems	Core systems that provide data, services and infrastructure: CRM, HRMS, job portals, email services, external databases and analytics platforms.

The diagram also lists the reasons for a hybrid approach: faster application development, flexible customisation, integration with existing systems, support for complex business logic and the ability to extend beyond platform limitations. The arrows between the layers indicate that information flows in both directions between the application, its extensions and the enterprise systems.

24140105

When Pro-Code Extension May Become Necessary

- ***Complexbusinesslogic:*** when businessrules andprocessing cannot be represented using standard workflows and rules.
- ***Specialized integrations:*** when the application must connect to legacy, HRMS or other systems for which suitable connectors are not available.
- ***Custom APIs:*** when internal services or interfaces have to be exposed or consumed in a way that the platform does not support directly.
- ***Advanced AI/ML:*** when intelligent features, such as resume screening, require custom models or machine-learning pipelines.
- ***High-performance processing:*** when large batches of data or real-time processing exceed what the platform can handle efficiently.
- ***Specialized user interfaces:*** when the required user experience goes beyond standard components.
- ***Complexenterprisedata processing:*** when large datasets and complex relationships require advanced data handling.
- ***Requirementsbeyond platform capabilities:*** when any needed functionality is simply not offered by the selected platform.

Case Study Analysis

develop the standard parts of the system rapidly: forms for job applications and applicant details, reports on application history, workflows for application processing and HR approval, notifications, and basic dashboards. These features correspond to the capabilities that low-code platforms are designed to provide.

As the organisation grows, the requirements become more demanding. A large number of applicants, applications and records raises performance and data-management concerns; integration with CRM, HRMS and job portals raises integration concerns; differing access needs and the handling of applicant personal data raise security and governance concerns; and advanced features such as intelligent resume screening may need external services. In such cases, extensions, integrations or custom development may be needed in addition to the low-code application. Low-code is therefore useful for rapidly developing standard business applications, whereas increasingly complex enterprise requirements may call for a hybrid approach, and the choice should be based on the requirements of the application.

Result

Thecase study demonstrated the major limitations that can arise when low-code platforms are applied to complex enterprise requirements, namely performance and scalability, complex business logic, system integration, customisation, security and governance, AI and advanced technology, vendor or platform dependency, and data management. It also showed how a hybrid low-code/pro-code approach can address requirements beyond standard low-code capabilities.

Conclusion

Low-codeplatforms offer speed and simplicity for standard business applications, but their limitations become more significant as enterprise complexity increases. For an application such as the Enterprise Job Application Management System, standard functionality can be built quickly using low-code, while demanding requirements in logic, integration, security, scale and advanced technology may need pro-code extensions. It is therefore important to understand the limitations of a platform before selecting a development approach, so that the chosen approach matches the requirements of the enterprise application.

24140105